

Smart OPD Registration Project

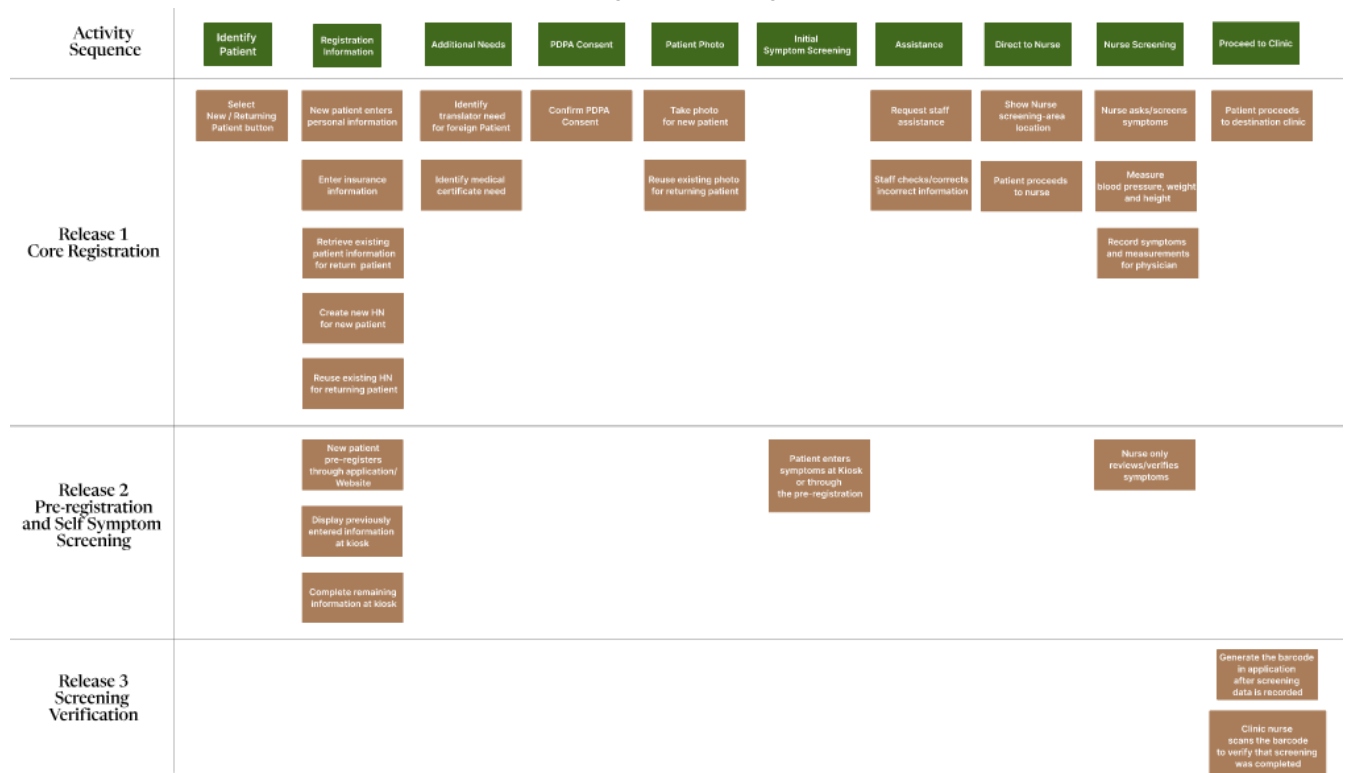
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Problem Statement & Business Benefits

Pain Point	Business Benefit
Long registration time and kiosk congestion	Pre-registration reduces kiosk usage time and congestion
Returning patients repeatedly enter existing information	Reusing existing patient data and HN to speed up registration process
Nurses spend time collecting basic symptom information	Self-reported symptoms reduce nurse workload and screening time
Patients may skip BP, weight, and height screening before proceed to the clinic	Screening verification helps ensure required steps are completed

User Story Map ([Link](#))

- **Phase 1 : Core Kiosk Registration**
Enable basic self-registration at Kiosk and proceed to nurse screening
- **Phase 2 : Pre-registration and Self-reported Symptoms**
Reduce Kiosk time and nurse workload by adding pre-registration and self symptom screening
- **Phase 3 : Screening Verification**
Ensure that patients complete required screening before visiting the clinic



User Story and Acceptance Criteria

Release 1

1. New Patient Registration

Story: As a new patient, I want to register at the kiosk so that I can start my hospital registration.

Acceptance Criteria:

Given I am a new patient

When I select "New Patient"

Then I can enter my personal and insurance information.

And the system creates a new HN.

2. Returning Patient Registration

Story: As a returning patient, I want to retrieve my existing information so that I do not need to enter it again.

Acceptance Criteria:

Given I have an existing HN

When I select "Returning Patient"

Then my information is retrieved from HIS

And my existing HN is used.

3. Foreign Patient Registration

Story: As a foreign patient, I want to register using my passport so that I can register without a Thai National ID.

Acceptance Criteria:

Given I am a foreign patient

When I provide my passport

Then I can continue the registration process.

4. Translator and Medical Certificate

Story: As a patient, I want to indicate if I need a translator or medical certificate.

Acceptance Criteria:

Given I am registering

When I select my additional needs

Then the system saves my selections.

5. PDPA Consent

Story: As a patient, I want to sign PDPA digitally at the kiosk.

Acceptance Criteria:

Given I have completed my registration information

When I reach the PDPA step

Then I can provide an electronic signature

6. Patient Photo

Story: As a patient, I want my photo to be used for my patient record.

Acceptance Criteria:

Given I reach the Patient Photo step

When I am a new patient, the kiosk takes my photo

Then the photo is saved

And an existing photo from HIS is used for a returning patient.

7. Request Assistance

Story: As a patient, I want to request assistance if I enter incorrect information.

Acceptance Criteria:

Given I need help at the kiosk

When I select "Request Assistance"

Then Registration Staff is notified.

8. Show Nurse- Screening area location

Story: As a Patient, I want to see the Nurse-Screening Area location after completing registration, so that I know where to go for the next step.

Acceptance Criteria:

Given the patient has successfully completed registration

When the HIS confirms that the visit/registration has been created

Then the kiosk displays the Nurse-Screening Area location and directs the patient to proceed there.

Release 2**9. Pre-registration**

Story: As a new patient, I want to pre-register through the application to reduce time at the kiosk.

Acceptance Criteria:

Given I am a new patient

When I pre-register through the application

Then my information is saved for use at the kiosk.

10. Show Pre-registration Information

Story: As a pre-registered patient, I want to see my existing information at the kiosk so that I do not enter it again.

Acceptance Criteria:

Given I have pre-registered

When I continue at the kiosk

Then my information is displayed

And I complete only missing information.

11. Screen Symptoms

Story: As a patient, I want to enter my initial symptoms to reduce the Screening Nurse's data-entry time.

Acceptance Criteria:

Given I am registering

When I enter my symptoms

Then they are saved for the Screening Nurse.

12. Verify Patient Symptoms

Story: As a Screening Nurse, I want to verify the symptoms from Kiosk before they are provided to the physician.

Acceptance Criteria:

Given the patient has entered symptoms

When I review and verify them

Then the verified symptoms are saved for the physician.

Release 3

13. Screening Barcode Verification

Story: As a Clinic Nurse, I want to scan the patient's barcode to confirm that screening has been completed.

Acceptance Criteria:

Given blood pressure, weight, and height are recorded

When I scan the patient's barcode

Then the system confirms the screening is completed.

Prioritization — MoSCoW

Priority	User Story	Reason
Must	New Patient Registration	It is a core function that allows new patients to start and complete the registration process.
Must	Returning Patient Registration	It prevents returning patients from re-entering existing information and allows them to reuse their existing HN.
Must	Create HN for New Patient	A new HN is required to create and identify a new patient's record in the hospital system.
Must	PDPA Consent	Consent is a required step in the registration process before patient registration can be completed.
Must	Direct Patient to Screening Area	Patients need clear instructions on where to go after registration to continue the patient journey.
Must	Record Screening Information	Symptoms, blood pressure, weight, and height need to be recorded so the physician can use them during consultation.
Should	Foreign Patient Registration	It expands the service to patients without a Thai National ID, but the first working flow can initially focus on Thai patients.
Should	Translator and Medical Certificate	These are important for patients who need them, but they are not required for every patient's core registration journey.
Should	Patient Photo	A patient photo supports the patient record, but registration can still function without kiosk photo capture in the first MVP.
Should	Request Assistance	It improves support and helps correct errors, but it is mainly needed for exception cases rather than the standard registration flow.
Could	Pre-registration	It can reduce kiosk usage time and congestion, but patients can still register directly at the kiosk without it.

Could	Foreign Patient Pre-registration	It improves convenience for foreign patients but depends on the Pre-registration capability and is not essential for the first release.
Could	Show Pre-registration Information	It reduces repeated data entry, but it is only needed after Pre-registration is introduced.
Could	Screen Symptoms	It can reduce the Screening Nurse's data-entry time, but the nurse can still collect symptoms directly in the first release.
Would	Screening Barcode Verification	The core screening process must work first. Barcode verification can be added later to help prevent patients from skipping required screening.

Iteration Plan

- **Release 1** — Focuses on the core end-to-end patient journey. The registration and screening process must work before additional convenience and verification features can be introduced.

Flow:

New / Returning Patient → Registration Information → HN → Translator / Medical Certificate → PDPA → Patient Photo → Request Assistance (if needed) → Screening Area → Nurse Screening → Save information for physician

- **Release 2** — Reduce the time patients spend at the kiosk and reduce the Screening Nurse's data-entry workload.

Flow:

Pre-registration through Application → New / Returning Patient → **Show Pre-registration Information** → **Complete Remaining Information (or skip the registration at Kiosk)** → HN → Translator / Medical Certificate → PDPA → Patient Photo → **Screen Symptoms** → Request Assistance (if needed) → Screening Area → **Nurse only Reviews & Verifies Symptoms** → Save information for physician

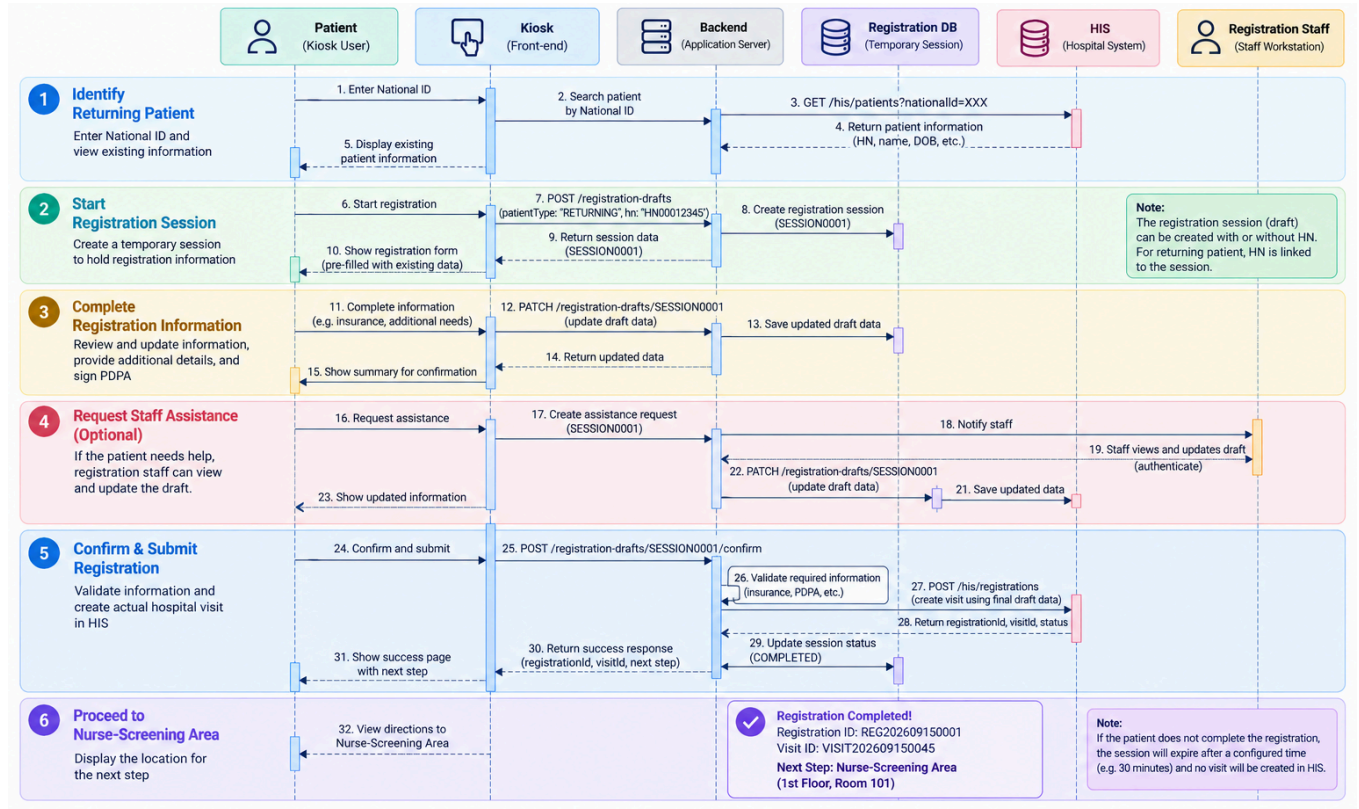
- **Release 3** — Ensure patients complete the required screening before proceeding to the destination clinic.

Flow:

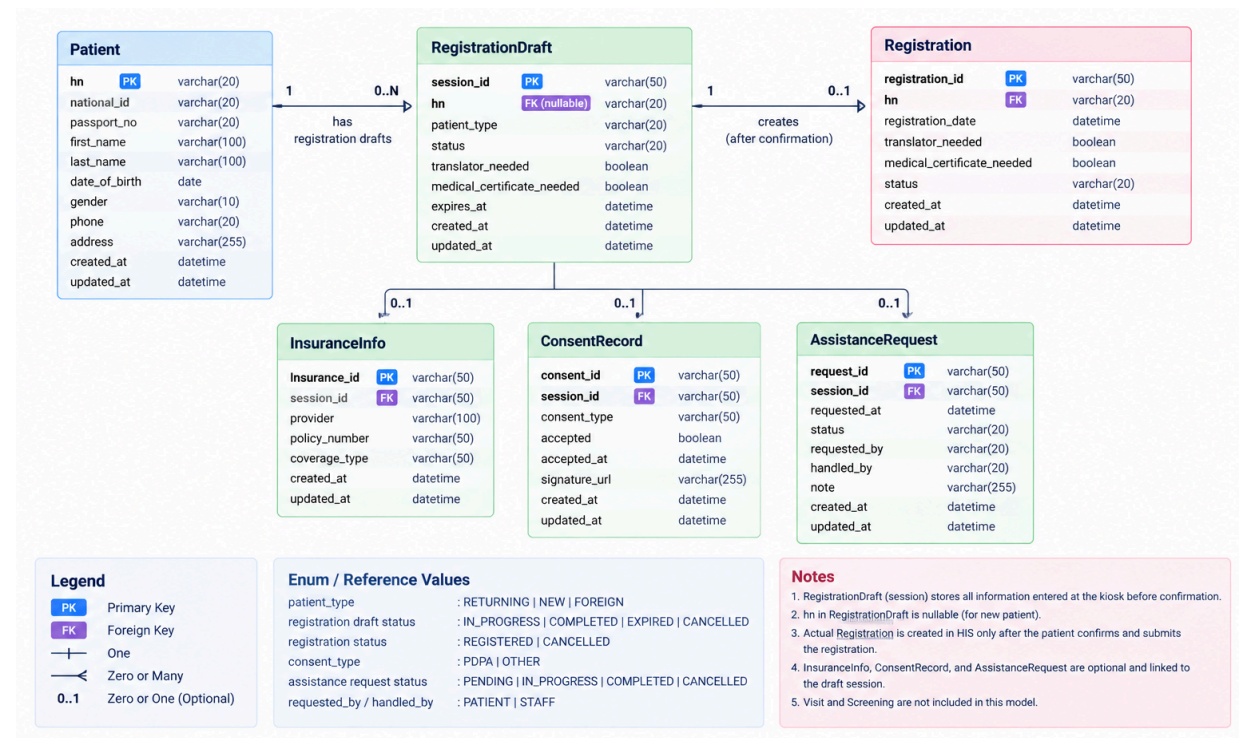
Pre-registration through Application → New / Returning Patient → Show Pre-registration Information → Complete Remaining Information → HN → Translator / Medical Certificate → PDPA → Patient Photo → Screen Symptoms → Request Assistance (if needed) → Screening Area → Nurse Reviews & Verifies Symptoms → Save information for physician → **Application Generates Barcode** → **Patient Goes to Destination Clinic** → **Clinic Nurse Scans Barcode** → **System Confirms Screening is Completed** → Patient Proceeds to the Next Step at the Clinic

Sequence Diagram

This sequence diagram shows the Returning Patient registration flow, from retrieving existing patient information and creating a temporary registration session to optional staff assistance and final confirmation. The actual registration in HIS is created only after the patient confirms and submits the registration, after which the kiosk directs the patient to the Nurse-Screening Area.



Data Model



API Specifications

Endpoint	Purpose	From → To	Example Request
POST/registration-drafts	Start a temporary registration session	Kiosk → Backend	POST /registration-drafts{ "patientType": "RETURNING", "hn": "HN00012345" }
GET/registration-drafts/{sessionId}	Retrieve the current registration draft	Kiosk → Backend	GET /registration-drafts/SESSION0001
PATCH/registration-drafts/{sessionId}	Update information in the temporary registration draft	Kiosk → Backend	PATCH /registration-drafts/SESSION0001 { "translatorNeeded": true, "medicalCertificateNeeded": false, "insurance": { "provider": "ABC Insurance", "policyNumber": "POL123456" } }
POST/registration-drafts/{sessionId}/assistance-requests	Request assistance from Registration Staff	Kiosk → Backend	POST /registration-drafts/SESSION0001 /assistance-requests{ "reason": "Incorrect insurance information" }
GET/registration-drafts/{sessionId}/assistance-requests	Allow authenticated staff to retrieve assistance requests	Staff Workstation → Backend	GET /registration-drafts/SESSION0001 /assistance-requestsAuthorization: Bearer <staff_access_token>
PATCH/registration-drafts/{sessionId}	Allow authenticated Registration Staff to correct draft information	Staff Workstation → Backend	PATCH /registration-drafts/SESSION0001 Authorization: Bearer <staff_access_token>{ "translatorNeeded": false, "medicalCertificateNeeded": true }

POST/registration-drafts/{sessionId}/confirm	Confirm and submit registration	Kiosk → Backend	POST /registration-drafts/SESSION0001/confirm
GET/his/patients?nationalId={nationalId}	Retrieve existing Returning Patient information and HN	Backend → HIS	GET /his/patients?nationalId=1103700123456
POST/his/registrations	Create the actual registration after the patient confirms the completed draft	Backend → HIS	POST /his/registrations { "hn": "HN00012345", "translatorNeeded": false, "medicalCertificateNeeded": true, "insurance": { "provider": "ABC Insurance", "policyNumber": "POL123456" } }

Edge Cases & Fall-backs

Edge Case	Fall-back
Returning patient cannot be found in HIS	Ask the patient to retry the identification or request assistance from Registration Staff.
Patient enters incorrect information	Use Request Assistance so Registration Staff can check and correct the information.
Required information is missing	The kiosk highlights the missing fields and prevents submission until completed.
HIS connection fails	Show an error message and direct the patient to the Registration Staff.
Patient photo cannot be captured	Allow the patient to retry or request assistance.
Foreign patient's passport cannot be identified	Ask the patient to re-enter the passport information or request staff assistance.
Patient cannot complete PDPA consent	Registration cannot be completed and staff assistance is provided.